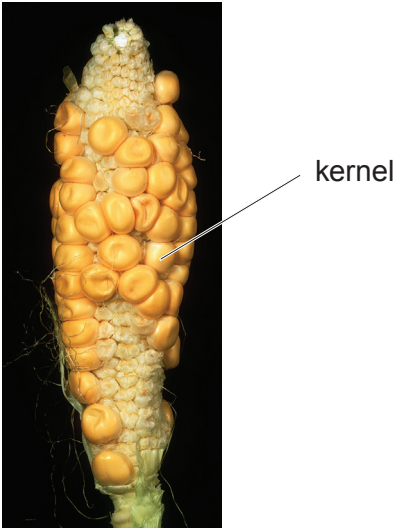


5. Angiosperms are the most successful plant group on earth, in part due to their reproductive strategies.

Following fertilisation in maize, fruits form which are called kernels. **Image 5.1** shows a poorly fertilised maize or corn cob with missing kernels.

Image 5.1



(a) (i) Describe double fertilisation in plants. [2]

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(ii) Explain why **only** the fertilised ovules develop into kernels. State the advantage to the plant of this. [2]

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Examiner
only

(b) Using **Images 5.2** and **5.3**, briefly describe how the following seeds are dispersed. [2]

Image 5.2



Dandelion

Image 5.3



Cocklebur

Dandelion

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Cocklebur

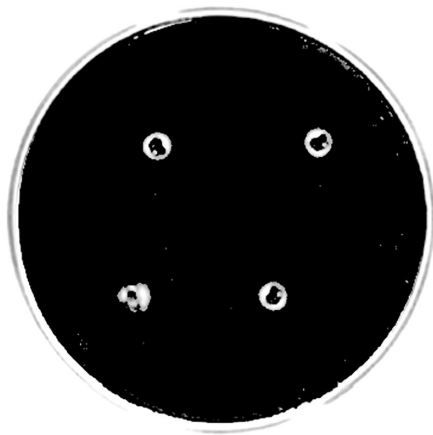
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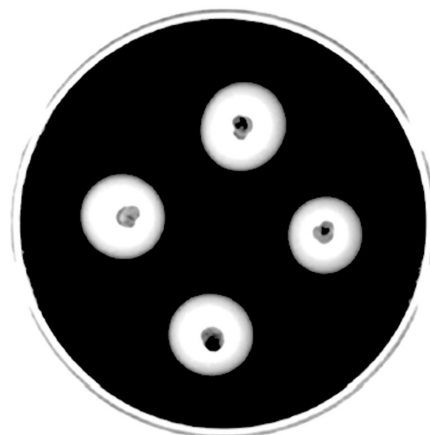


- (c) An experiment was carried out to investigate the effect of the embryo on controlling germination in barley grains. The method is given below.
- Half-grains were incubated on the surface of starch-agar gel.
 - After 48 hours, the gel was washed with iodine-potassium iodide which reacts with starch to form a blue-black colour.
 - A clear circle, or halo, surrounding the half-grain indicates the digestion of starch by α -amylase.
 - The results are shown in **Image 5.4**. The control plate (left) contained four half-grains with no embryo present. The plate on the right contained half-grains with embryo intact.

Image 5.4



Control plate: no embryos present



Embryos present



Examiner
only

(i) State the following variables in this experiment: [2]

Independent variable

Dependent variable

(ii) State the conclusion you can draw from the experimental results shown in **Image 5.4**. Use your knowledge of plant hormones to explain your answer. [3]

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